Swesan Pathmanathan

Canadian Citizen

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Education

McMaster University Hamilton, ON
Bachelor of Engineering, Software Engineering (Co-op) *Sept 2021 – Apr 2026*

Technical Skills

Languages: Python, C++, C, TypeScript, SQL, Java
Databases: PostgreSQL, IBM DB2
AI Tools: Cursor, GitHub Copilot
Frameworks & Libraries: React Native, Node.js, PyTorch, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib
Systems & Infrastructure: Linux, Windows, Docker, GitHub Actions, ROCm, BMC, BIOS/UEFI
Developer Tools: Git, Bash, Quartus

Work Experience

Advanced Micro Devices (AMD) | Python, C/C++, ROCm, Linux, GitHub, Docker, post-silicon GPU validation, firmware, BMC, BIOS/UEFI **Markham, ON**
Datacenter GPU System Integration Engineer (Co-op) *May 2024 – Aug 2025*

- Acted as rotating **test lead** for MI300X, MI325X, and MI350X post-silicon GPU validation, coordinating **100+ runs weekly** on **Linux** and **Windows** across HPC and ML workload gates.
- Coordinated feature enablement with firmware, validation, and infrastructure partners across MI300X, MI325X, and MI350X profiling and workload paths, keeping cross-team integration milestones aligned for release cycles.
- Triaged and reproduced ML workload failures across **Docker** images and the **ROCm** runtime on MI300X, MI325X, and MI350X systems, narrowing software versus hardware root cause before reruns.
- Debugged **BMC** and **BIOS/UEFI** issues across MI300X, MI325X, and MI350X racks; documented flash and bring-up steps reused on **20+ rack bring-ups**, **cutting repeat setup clarifications by about 40%**.
- Extended **Python** automation and an internal workload launcher with **GitHub**-tracked scripts, shrinking diagnostic setup from **about 20 minutes to under 5 minutes** per run through reusable launch paths.
- Triaged defects across vendor **GPU** stacks including ROCm, firmware, **C/C++**, and interconnects using logs; added **repeatable test scripts** so every handoff shipped the same repro bundle, which **cut reopened and duplicate defects on my queue by about 30%** by eliminating one-off retests.

Projects

Hitchly Ride Share Application | Node.js, PostgreSQL, Docker *Sept 2025 – Apr 2026*

- Implemented CI and release checks with GitHub Actions and **Docker** for linting, tests, and pull request validation, strengthening production software quality gates for each backend change.
- Built and integrated backend **APIs** in **Node.js** with a **Docker**-containerized **PostgreSQL** service, supporting client-server workflows and feature integration across the application stack.
- Integrated mobile trip-booking flows with backend APIs across **React Native**, **TypeScript**, and **Node.js** services, improving end-to-end feature delivery across client and server components.

Gym App Platform | Next.js 16, React 19, TypeScript, Supabase, Postgres *May 2026*

- Built a full-stack gym tracking platform with **Next.js 16** App Router and **React 19**, using **TypeScript** for shared client/server types across workout, history, and progress flows.
- Implemented backend data access with **Supabase** and **Postgres**, including SQL migration workflows for schema evolution and reproducible environment setup.
- Developed analytics views with **Recharts** and date-based trend rollups with **date-fns**, and prepared deployment-ready configuration targeting **Vercel** with basic PWA add-to-home-screen support.

Linux Log Normalization and Token Frequency Analysis | Python, NumPy, pandas *Jan 2026*

- Developed a configurable text normalization pipeline in **Python** for processing raw **Linux logs**, supporting tokenization, stemming, lemmatization, and digit filtering.
- Performed token frequency computation and statistical aggregation using **NumPy** and **pandas** to analyze distributional patterns in **Linux logs**.

Ontario Hospital Association Database Design | SQL, IBM DB2 *Sept 2025*

- Designed a relational schema for a hospital information system modeling patients, records, employees, and clinical workflows to support consistent operational queries.
- Authored SQL scripts implementing primary keys, foreign keys, and referential integrity constraints to ensure consistency across patient and record data.
- Developed a full database creation script compatible with **IBM DB2** CAS server environments for reproducible deployment.

SufSort Sorting System | C, Bash, Linux *Nov 2025*

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.
- Measured runtime behavior across multiple input sizes and used test logs to compare sorting throughput, making algorithm tradeoffs visible for larger file workloads.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy *Feb 2026*

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

Pneumonia Detection CNN | Python, TensorFlow/Keras, NumPy, pandas, scikit-learn *Jan 2026*

- Built a convolutional neural network classifier in **Python** using **TensorFlow/Keras** to detect pneumonia from chest X-ray images.
- Applied data preprocessing, dataset splitting, and augmentation using **NumPy** and **pandas** to improve model generalization on imbalanced clinical datasets.
- Evaluated model performance using **accuracy**, **precision**, **recall**, confusion matrices, and visualization tools to guide architecture and hyperparameter tuning.

Breast Cancer Prediction Neural Network | Python, PyTorch, NumPy, scikit-learn *Sept 2025 – Dec 2025*

- Built a feedforward neural network with **PyTorch**, using **NumPy** and **scikit-learn** preprocessing and stratified splits, to classify malignant and benign breast cancer samples from tabular clinical datasets.
- Applied regularization techniques including **dropout** and **L1/L2** penalties to improve model stability and reduce overfitting.

ALU Design with Sequential Circuits | Verilog, Quartus *Sept 2025*

- Structured a synthesizable ALU datapath from small **Verilog** blocks and checked behavior block-by-block before full-chip stitching.
- Coded deterministic **Verilog testbench** stimulus plus self-checks, exercised ALU arithmetic and logic modes over a scripted input matrix, and used **Quartus** waveform debug until every planned functional regression passed.

Terrain Generation System | Java, Maven, Git, JTS *2024*

- Developed a software tool in **Java** for generating and visualizing 2D terrain meshes for simulation and visualization workflows.
- Modeled vertices, edges, and polygons using the **JTS** geometry library to represent terrain features and topology.
- Implemented a command-line interface to configure generation parameters, run batch experiments, and visualize output meshes.

Relevant Courses

Performance Analysis Tools and Methods, Concurrent Systems and Synchronization, Computer Networks Security, Database Systems

Hobbies & Interests

Gaming, Hardware/Software Tinkering, Gym.